

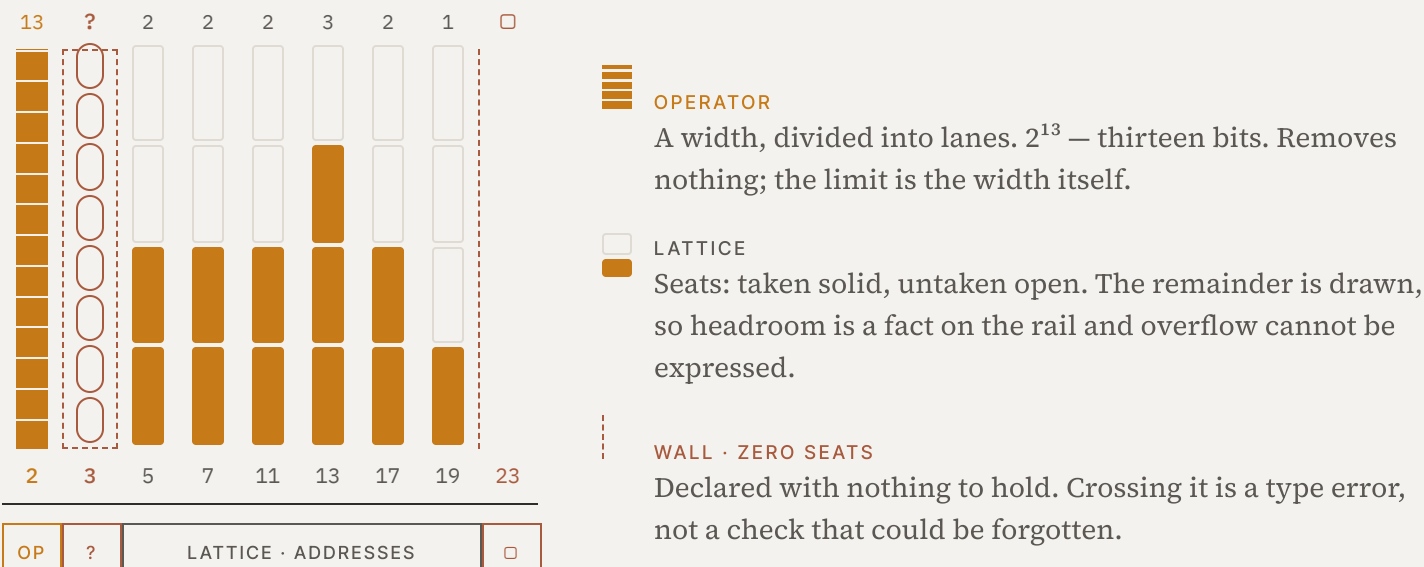
The abacus reads depth, not magnitude.

One rod per prime, one bead per power. The decimal form is the number's *name*; the rod profile is its *structure*. Two coordinates, not an implication.

THE TYPED RAIL

Capacity tells operator from lattice.

A lattice-bound quantity exists by **removal** from a declared stack — it spends address space, and what it did not spend stays on the rail as open seats. An operator removes nothing: it is bounded by its own bit width, acts only on the stack it possesses, and takes its structure from the other rods rather than declaring it. Two capacities, two grammars, and they are not interchangeable — which is the whole reason to draw them differently.



Rod 3 is not filed. §2 places 3^8 in the fibre as the trit register, *instantiated, never stored* — which reads operator-side. But an operator's limit is stated as a *bit* width, a 2-power, and 3^8 is not one. Either the operator bound admits non-binary widths, or the trit register is lattice-bound and the fibre spans both classes.

It is drawn refused rather than defaulted. Filing an operator inside the space it operates on is the documented failure mode, four times over; a rod guessed here would be the fifth, and would look identical to a rod that was known.

Nothing on this rail permits an operator to be filed inside the space it operates on — the two grammars cannot be mistaken for one another at a glance, which is the only place that error was ever available.

CONSTRUCTING THE ODD RODS

Only two is a position. The rest are joined.

The machine's vocabulary is positions and shifts, so multiplying by two is wiring and costs nothing. Every odd rod has to be **joined** out of positions, and the identity that licenses the join — $3 = 4 - 1$, $5 = 4 + 1$, $7 = 8 - 1$, $17 = 16 + 1$ — is arithmetic, which sits outside the machine. That is the harder part: each odd prime becomes a construction rather than an instruction. The payoff is that nothing here needs a multiplier.

Cost is counted in add and subtract steps — one fewer than the positions used. Shifts are wiring and are not counted. Signed forms are used throughout because a subtraction is the same price as an addition, which is what makes 7 and 23 cheap.

ROD	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	RECIPE	OPS	DEPTH	TOTAL
2	•	•	•	+	•	x << 1 position · wiring	free	13	0
3	•	•	+	•	−	(x << 2) − x class undeclared	1	8	8

5	. . + . +	$(x \ll 2) + x$	1	2	2
7	. + . . -	$(x \ll 3) - x$	1	2	2
11	. + . + +	$(x \ll 3) + (x \ll 1) + x$	2	2	4
13	. + + . +	$(x \ll 3) + (x \ll 2) + x$	2	3	6
17	+ . . . +	$(x \ll 4) + x$	1	2	2
19	+ . . + +	$(x \ll 4) + (x \ll 1) + x$	2	1	2
23	+ + . . -	$(x \ll 4) + (x \ll 3) - x$ <i>constructible · undeclared</i>	2	□	—

THE BASE, CONSTRUCTED

18

add / subtract steps for $5^2 \cdot 7^2 \cdot 11^2 \cdot 13^3 \cdot 17^2 \cdot 19$ — zero multiplies, zero divides, no float. Shifts uncounted.

THE TRIT REGISTER

8

steps for 3^8 if it is constructed rather than held — contingent on rod 3's class, which is undeclared.

WEIGHT TWO

3 · 5 · 7 · 17 · 31

Every prime to 71 reachable from two positions. Measured over that range only; no claim past it.

KILLED HYPOTHESIS · RECORDED, NOT DROPPED

That 23 is the wall because it is the first prime here needing four positions. It is not: $23 = 16 + 8 - 1$, three positions — the same weight as 11, 13 and 19, and therefore the same cost at equal depth. Signing is what buys that: unsigned, 10111 is four positions and three steps; signed, it is three positions and two. So the saving is against 23's own unsigned form and against nothing else. Construction cost does not explain the wall, and the wall's row above shows an ordinary recipe with no seats declared to run it in. The arithmetic was never what forbade it.

SUPPORT BEAD

● depth 1 · the radical

Present on every rod the integer touches. Read alone, the bottom row is $\text{rad}(n)$.

THE PLATE

9.610... [hover for the name](#)

DEPTH BEADS

● every power above one

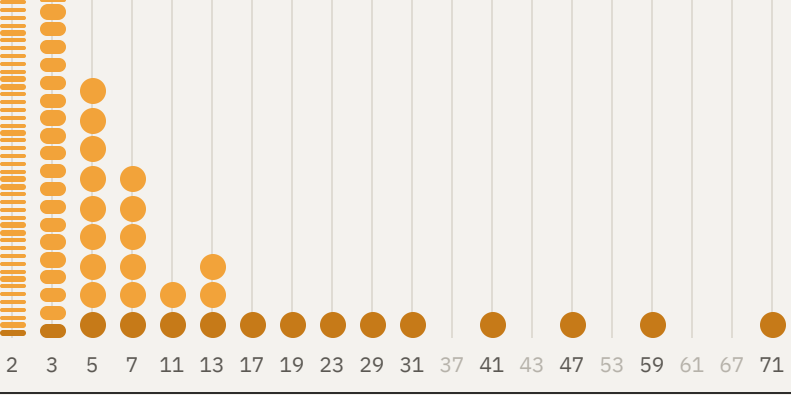
The stack is the exponent, unary and literal. Height is depth, not value.

THE WALL

□ [absent by design](#)

A dashed rod is drawn, never omitted. The empty rod is the statement.

54 digits · exact · |M|



BASE-10 FORM

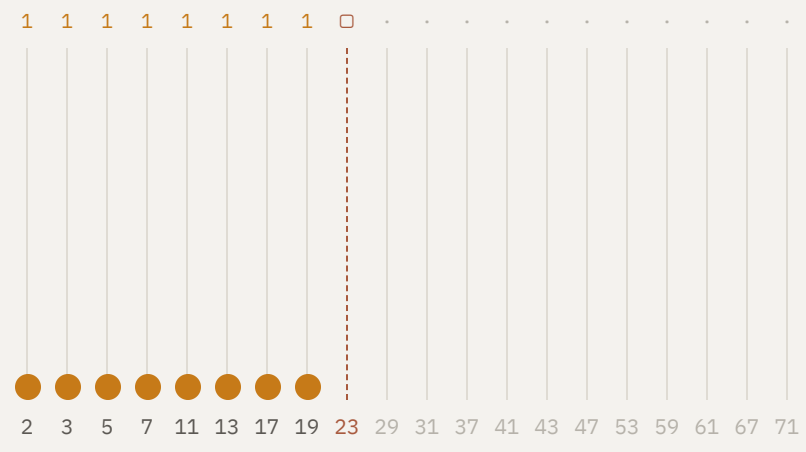
8.0801742479... ×
10⁵³

54 digits – past reading.
The mantissa is truncated at
ten places, not rounded; the
exact integer is the prime
signature above.

Depth on exactly {2,3,5,7,11,13}, flat from 17 up — the two-sided bitmap
at the top of the hierarchy, and it reads off the silhouette without a
table. Against rokujiku it grips on every rod but one: 17¹ against 17², a
shallower hold.

rad(rokujiku) = 19# ✓ SQUAREFREE

2·3·5·7·11·13·17·19 · divides 773#



MAGNITUDE

9,699,690

7 digits · exact · squarefree

SHOWN WHOLE

9,699,690

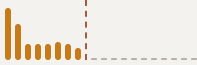
Seven digits. No reduction
needed, so none is applied.

Every rod at depth one. A squarefree period cannot contain a modulus that
has depth — on the instrument that stops being a claim and becomes a
silhouette: a flat row of single beads has nowhere to put a stack. Seven
digits, so the plate carries the number whole; nothing is reduced that does
not need reducing.

Small enough to set inline.

At this size the beads cannot be counted, so the comb surrenders countability and keeps the silhouette. A bar's height is a fixed presence floor plus one unit per power — affine, not proportional, because a proportional scale put depth one within a pixel of the absence mark and the radical's row read as an empty row. The floor is constant, so differences in depth stay exactly proportional and the scale remains shared and absolute: a comb is never normalised to itself. A bar deeper than the scale runs full height and carries a terra-cotta cap above it, admitting the clip; an untouched prime keeps a faint mark so an absence still registers.

Set in a sentence it behaves as a glyph: the kernel  falls steeply from two to three and runs level from five up, and that much is legible mid-line without a figure. Where the count itself is load-bearing, use the full rail.

MODULUS	COMB	SIGNATURE	MAGNITUDE
rokujiku		$2^{1^3} \cdot 3^8 \cdot 5^2 \cdot 7^2 \cdot 11^2 \cdot 13^3 \cdot 17^2 \cdot 19$	$9.610... \times 10^{19}$
The Monster		$2^{4^6} \cdot 3^{2^0} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71$	$8.080... \times 10^{5^3}$
rad · 19#		$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19$	9,699,690

Three rows on one scale: the kernel's weight at the bottom, the Monster's two capped bars standing above it, and the radical's even row along the presence floor. The comparison that needed three full rails fits in a ledger column.

An instrument built once, so that a twenty-digit integer never again has to be read as a quantity.

